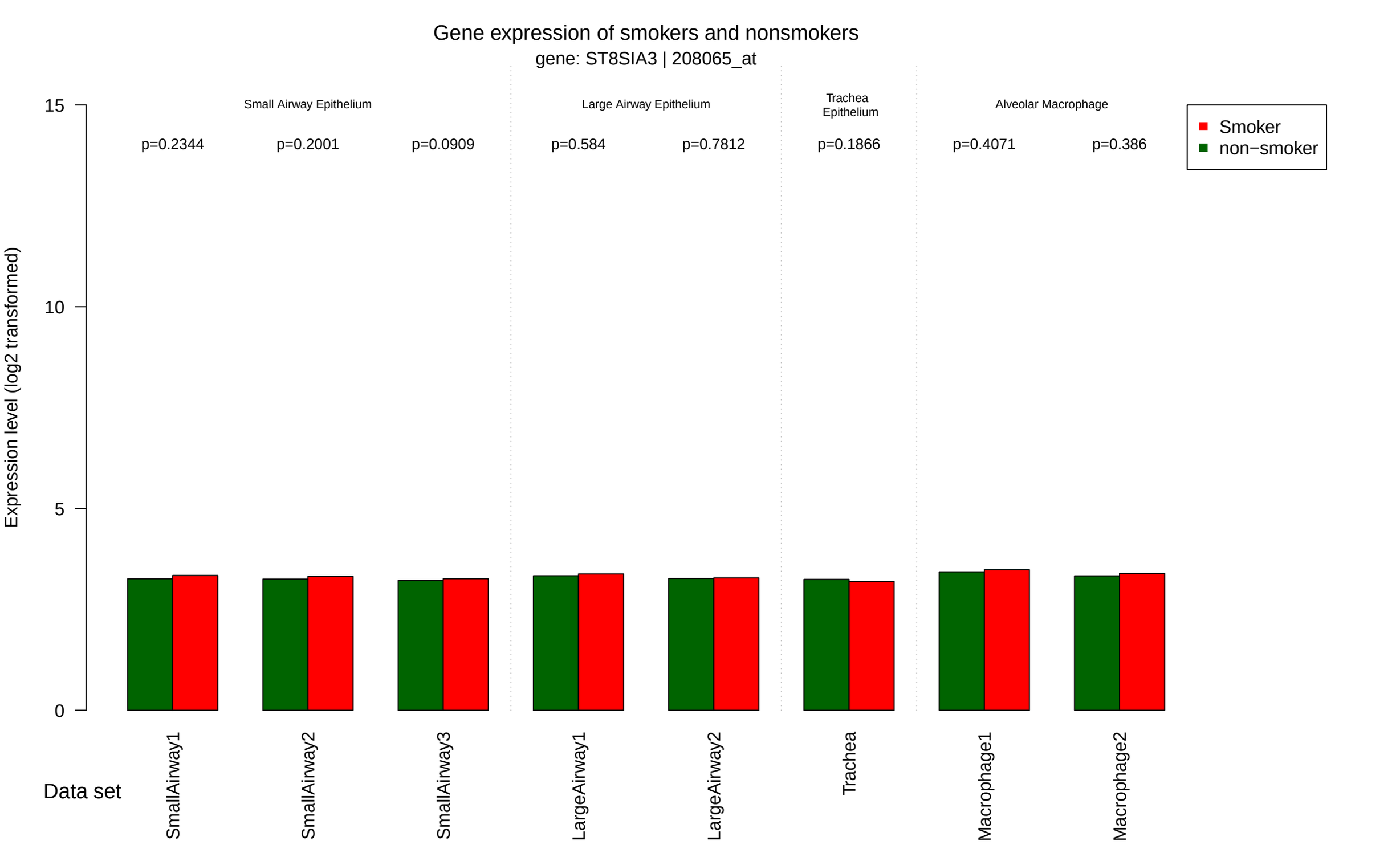
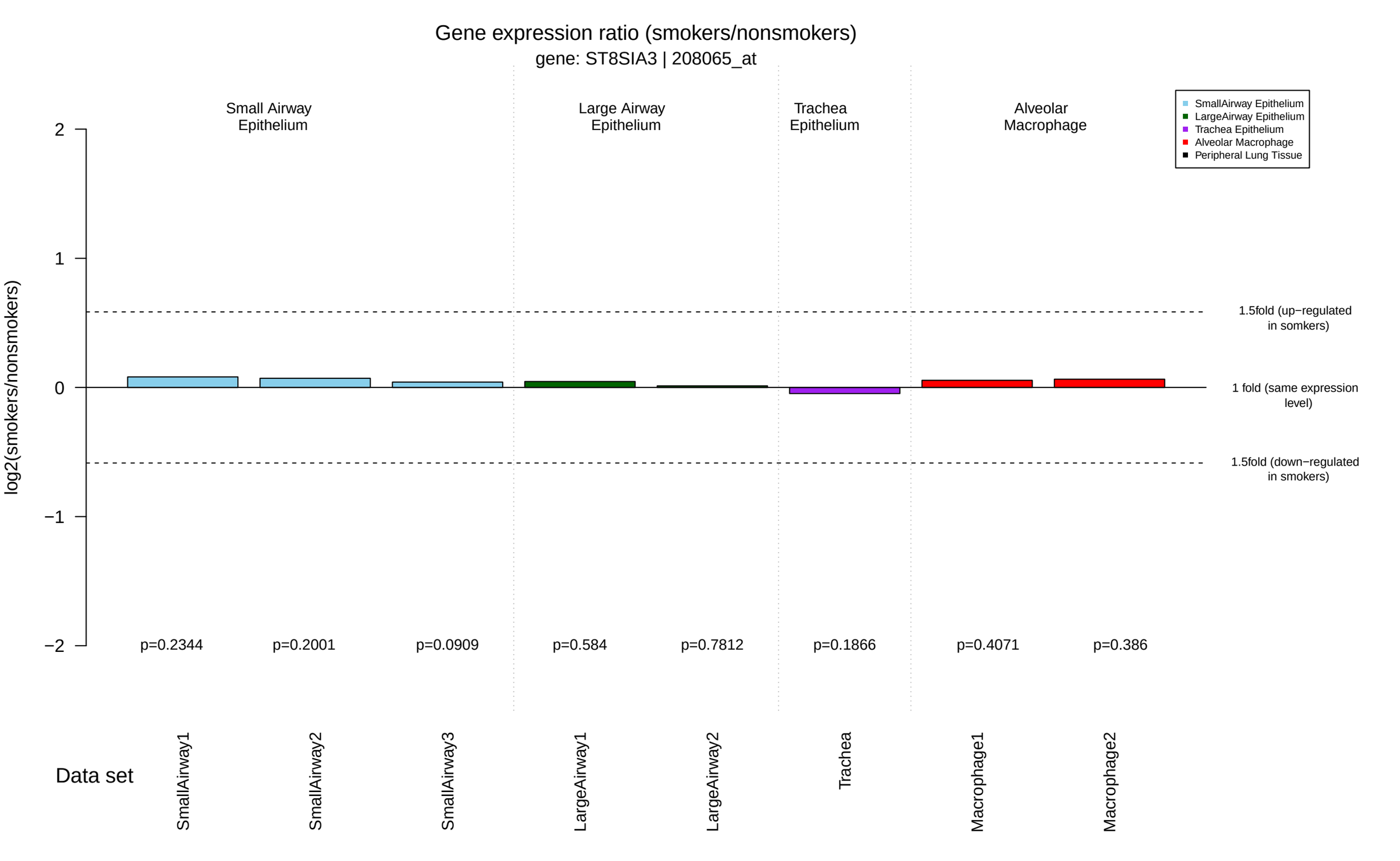
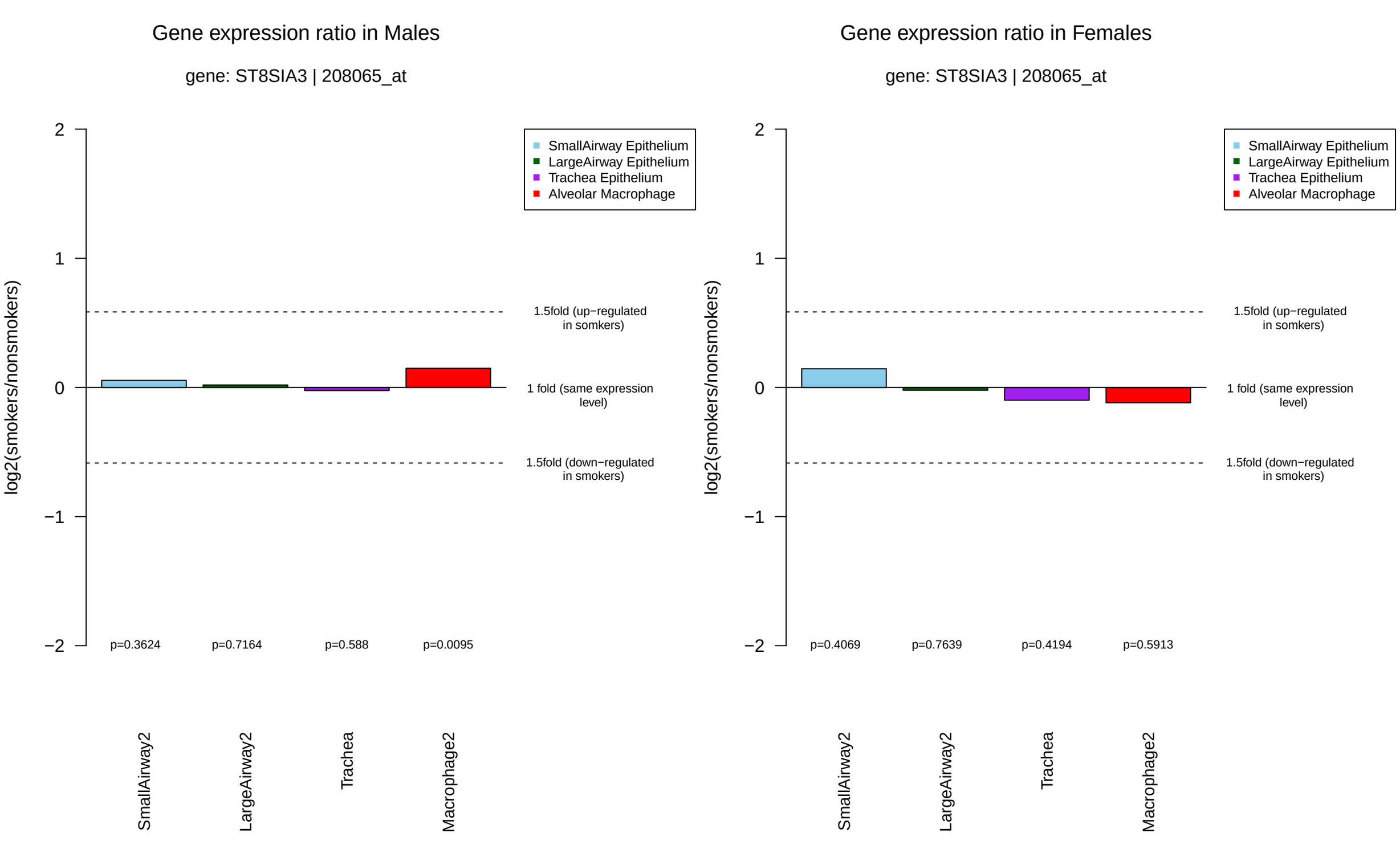
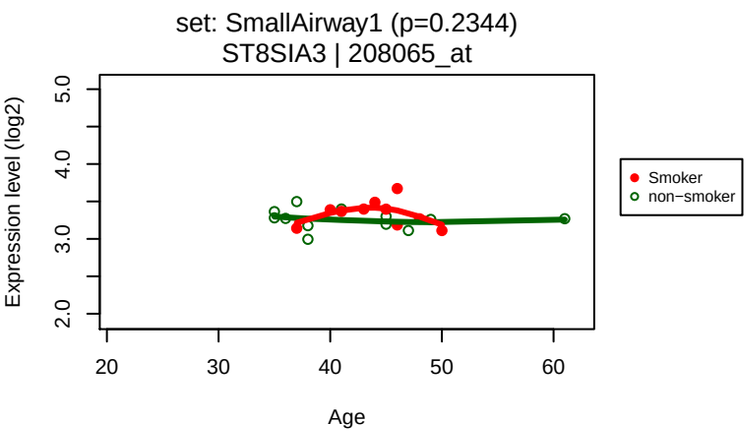
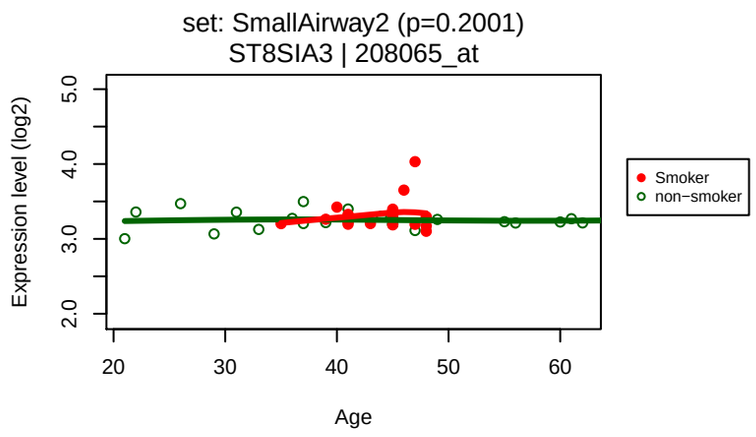
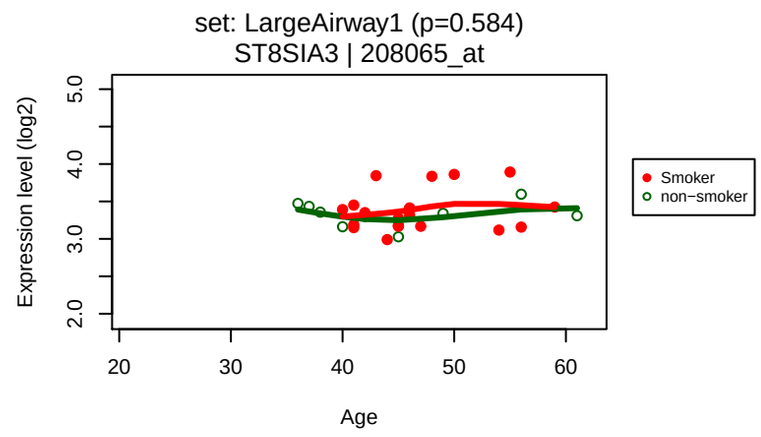

Gene expression of smokers and nonsmokers
gene: ST8SIA3 | 208065_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2344
p=0.2001
p=0.0909
p=0.584
p=0.7812
p=0.1866
p=0.4071
p=0.386
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

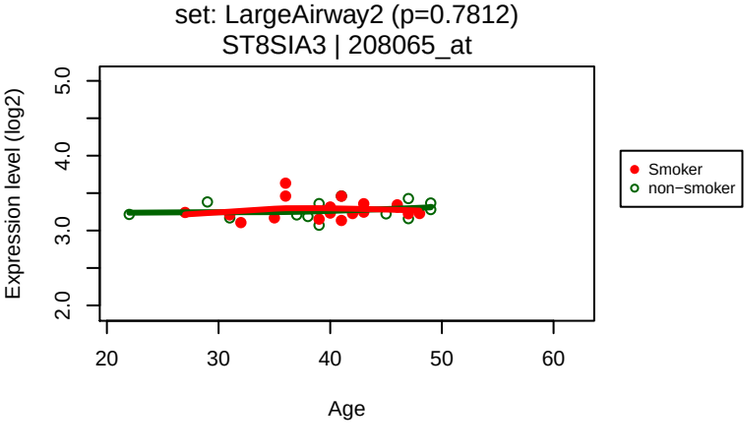

Gene expression ratio (smokers/nonsmokers)
gene: ST8SIA3 | 208065_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2344
p=0.2001
p=0.0909
p=0.584
p=0.7812
p=0.1866
p=0.4071
p=0.386
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

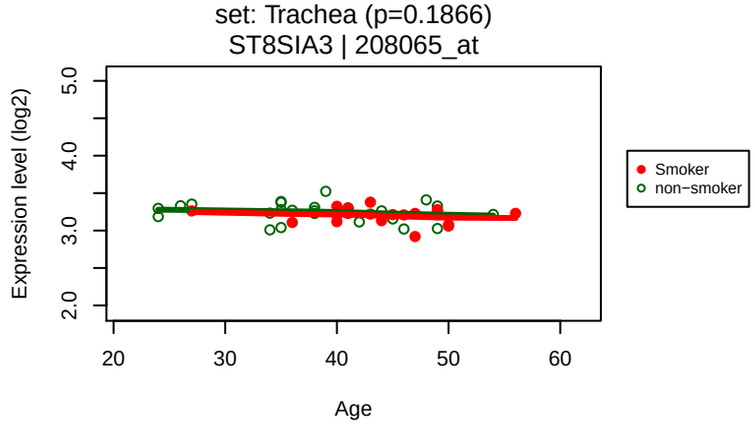

Gene expression ratio in Males
gene: ST8SIA3 | 208065_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3624
p=0.7164
p=0.588
p=0.0095
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST8SIA3 | 208065_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4069
p=0.7639
p=0.4194
p=0.5913
SmallAirway2
LargeAirway2
Trachea
Macrophage2

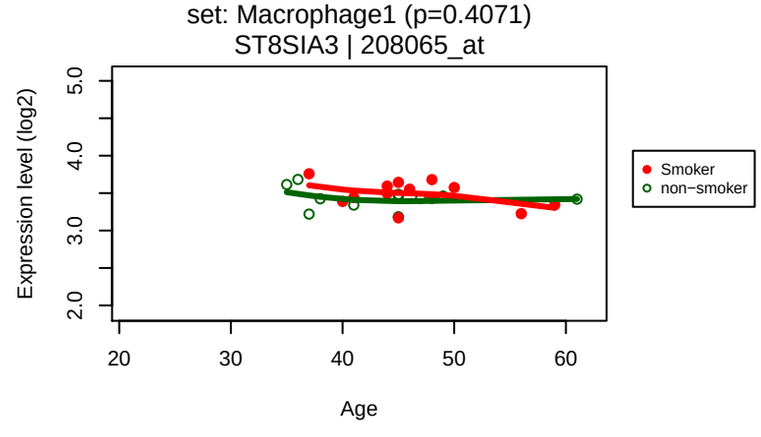

set: SmallAirway1 (p=0.2344)
ST8SIA3 | 208065_at
Expression level (log2)
Age
Smoker
non−smoker

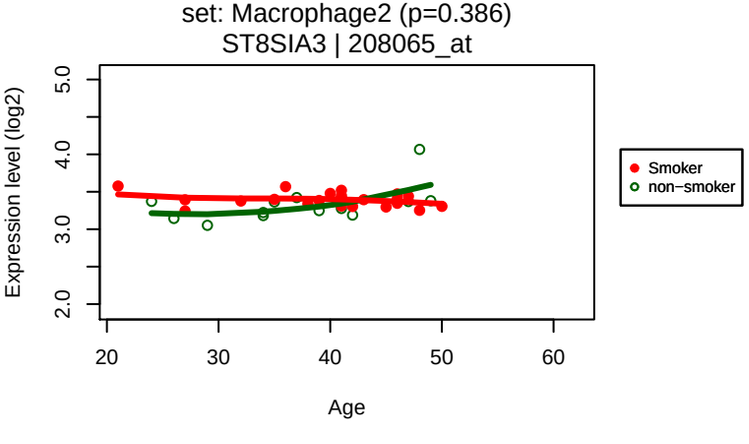

set: SmallAirway2 (p=0.2001)
ST8SIA3 | 208065_at
Expression level (log2)
Age
Smoker
non−smoker

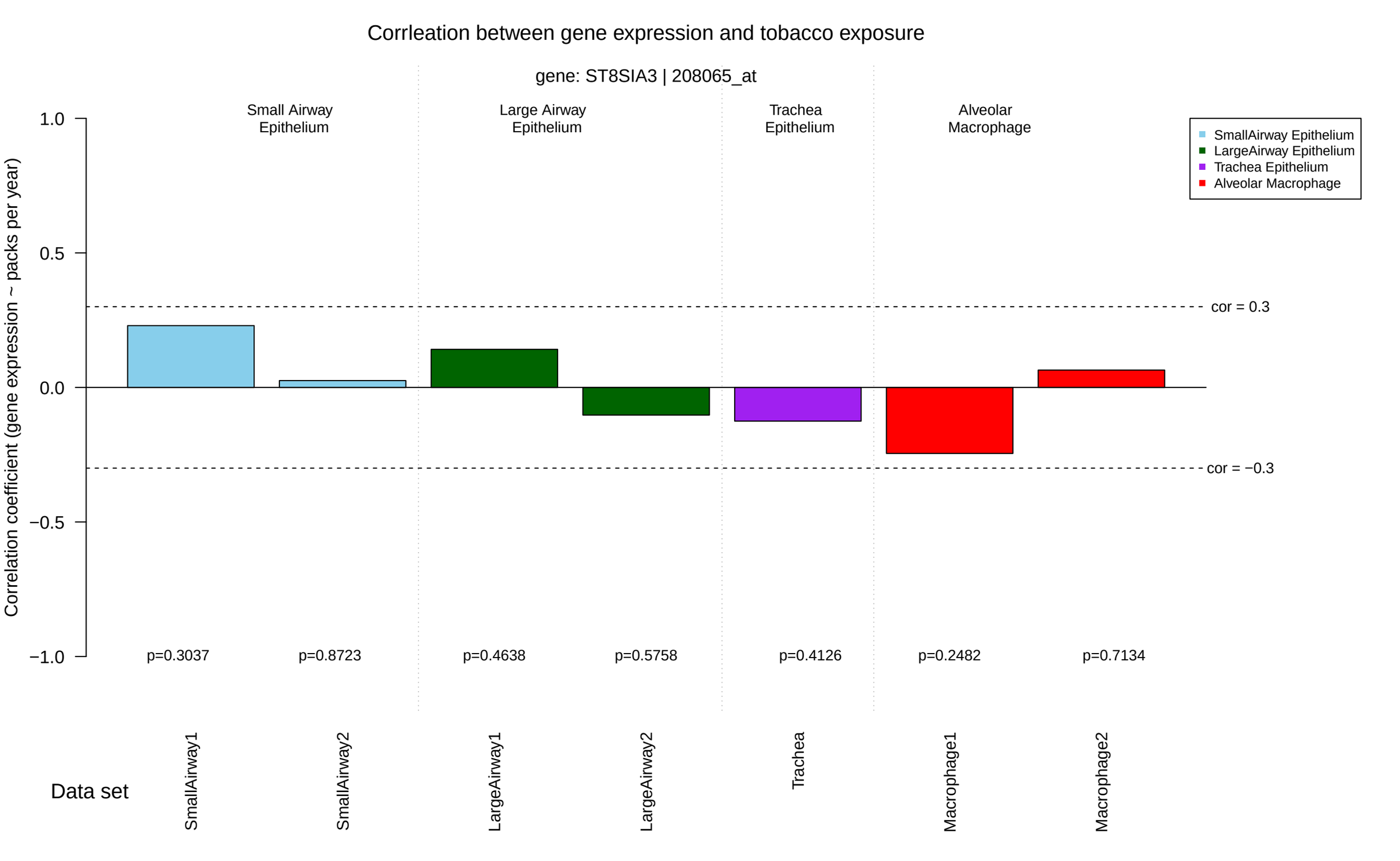

set: LargeAirway1 (p=0.584)
ST8SIA3 | 208065_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7812)
ST8SIA3 | 208065_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1866)
ST8SIA3 | 208065_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4071)
ST8SIA3 | 208065_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.386)
ST8SIA3 | 208065_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST8SIA3 | 208065_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3037
p=0.8723
p=0.4638
p=0.5758
p=0.4126
p=0.2482
p=0.7134
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2